

Introduction

Magic: The Gathering (MTG) is the first and most popular trading card game created in 1993 by Wizards of the Coast, the same company that created Dungeons & Dragons. Players play lands and use mana derived from played lands to cast spells and combat with other players. Typically, a player wins when all other opponent's health is reduced to zero although there are alternative winning conditions that exist. One project member plays MTG at least once a week with others and stays updated on the latest released cards that can optimize their decks and improve their strategies. Tools like EDHrec, Scryfall, and Archidekt are used by several MTG players to build and test their decks prior to purchasing the cards off the market.

For this project, 3 different methods were evaluated using both supervised and unsupervised learning algorithms. Ridge Regression, K-Nearest Neighbors Regression, and Random Forest Regression were analyzed due to all three supervised models having vastly different algorithms. The R-squared score, Mean Squared Error, and Mean Absolute Error were compared to understand which model fits the creature and non-creature card dataset. For the unsupervised methods, the U-MAP with K-Means Clustering and DBSCAN models were analyzed to visualize clusters of similar creature cards and artwork images. The best model would be identified by evaluating silhouette scores (specifically the Davies-Bouldin index and Calinski-Harabasz index). With optimal clusters, identifying patterns of any aesthetic differences of the artwork or price values within a cluster of similar cards.

Both analyses will evaluate specific card elements and determine if a model can identify a correlation between specific elements and the market price value of MTG cards. By understanding the impacts derived from the project results, the price of new cards could be estimated prior to a new release or reprint of an existing card. In addition, a player can estimate the approximate cost of a deck by inputting a list of cards into the model. Inferences can also be identified as to what strategies may be popular and sought after by players due to the high cost. The price of cards change everyday, so this model could be updated with the latest price dataset to understand what card elements are important at any given time.

Related work

1st Project - "Best MTG Deck For You" by Lorenzo and Yang Yunyi

Lorenzo and Yang Yunyi created a GitHub repository and hosted a website, called the '[Best-MTG-Deck](#)' and '[Best MTG Deck For You](#)' respectively, that evaluates a deck of cards and outputs approximately how much a user should pay to create their deck based on the latest market pricing. According to the website, the price data utilized for the 'Best-MTG-Deck' model is derived from TCGPlayer's API. Our project will not only be analyzing the market price value, we will also be focusing on aesthetic value by observing artwork, in-game value integrating card elements into the model, and popularity value by referencing rankings and ratings acquired from deck building sites like EDHrec.

2nd project - "I Built My Own EDHREC... and It Actually Works" YouTube Video by GameFreakSA

GameFreakSA was able to acquire over 650,000 deck lists using an API to create a new recommender system that would suggest specific MTG cards based on a list of cards. This recommender system (found on website [recommender.cards](#)) is different then the EDHREC recommender given it omits cards that synergizes with all decks, mining out cards that don't belong in the deck, and understanding the root strategy of the input deck. Like this recommender system, our project will be looking at similar variables and text data, and will be looking into identifying cards with similar attributes to compare. However, this project's focus will be concentrating on market price and predicting cost instead of deck building.

3rd project - "AI Education Matters: Data Science and Machine Learning with Magic: The Gathering" by Todd W. Neller

Neller utilized the same MTGJSON dataset that was used for the supervised learning portion of this project to estimate the total mana cost to play a creature card that contains the word "flying" based on if the card includes the word "flying", the power, and the toughness. Our project will be observing the market value price instead of the in-game cost, and will be observing several other variables of the card including the text, the power, and toughness.

Data Sources

We used two different datasets, one for supervised and one for unsupervised. The dataset we used for our supervised learning models were .csv files from [MTGJSON.com](https://mtgjson.com/), an open-source project collecting daily information from MTG's own Gatherer and the Scryfall API, and price information from TCGPlayer API. The data we used for unsupervised learning were .json files from [Scryfall.com](https://scryfall.com/)'s API, containing similar data to MTGJSON above, but also includes links to the artwork, set information, and scryfall webpage of the card. Both datasets contained cards dating from the 1990s (when Magic: the Gathering was released) to the 2020s. We downloaded and accessed these files from January 3, 2026.

Supervised Learning - Feature Engineering

We removed variables that tracked IDs in various databases and checked for the presence of some keywords and subtypes and added them as boolean features. We also condensed the data to remove outliers from the prices - some were in the ten-thousands and some cost only cents. We then appended one-hot-encoded dataframes for categorical features like color identity. We also excised rows with NaN values in EDHREC rank - as it is a key variable that is important to determining price, and not having a rank usually means a card is not in the EDHREC database, which also means not having a saltiness ranking from EDHREC (lack of saltiness rating with presence of a rank usually means the card hasn't been played enough).

Our target variable was the card's market price in USD. See Appendix A for our final features.

Supervised Learning - Models Analyzed

We used three different models - ridge regression, random forest regression, and k-nearest neighbors regression - to examine the effects of a card's features on predicting market price.

The k-nearest neighbors model averages similar data points (cards) from the inputted dataset. Hyperparameter tuning for the KNN regression model was done through creating an elbow graph for multiple different values of k plus picking the best value from looking at the resulting graph and a GridSearchCV.

We also used Term Frequency-Inverse Document Frequency (TF-IDF) on the cards' ability text to extract frequently used words that have a gameplay connotation in order to further look at whether they influenced card prices, and whether there were any similarities in attributes or card mana (a resource that represents the cost to "use" a card).

Random forest regression is an ensemble learning method that averages the predictions of multiple trees trained on random subsets of data to produce a more stable and accurate prediction. While individual decision trees are more prone to high variance and overfitting, especially when modeling complex data, the aggregation of many decision trees improves generalization performance. We chose this model because it effectively captures non-linear relationships and complex feature interactions without requiring extensive data preprocessing such as normalization or scaling. Random forest models also provide feature importance scores, which helps identify which features contribute the most to prediction.

We trained and tuned the model using a grid search with 5-fold cross validation and the hyperparameters 'n_estimators', 'max_depth', 'max_features', and 'min_samples_leaf'. The 'n_estimators' parameter defines the number of decision trees to build within the forest. More trees generally improve model performance by reducing variance and helping the model generalize, but can significantly increase computational cost and time for training and prediction. The 'max_depth' parameter controls the maximum number of levels in each decision tree. A high 'max_depth' allows the tree to grow deep and capture complex patterns, but may be more prone to overfitting while a low 'max_depth' makes the model simpler, but may lead to underfitting. The 'max_features' controls the number of features the model looks at when determining the best split, with lower values reducing overfitting by increasing randomness.

among trees. A recommended approach is to use the square root of the total features. The 'min_samples_leaf' specifies the number of samples required to be at a leaf node. It is also used to control the complexity of trees and prevent overfitting. A single decision tree from the ensemble is shown in Appendix B1.

Ridge regression is a linear regression that applies an L2 regularization parameter, lambda, to the squared values of the coefficients within the loss function. As the lambda value increases, the large coefficients will increase significantly while the coefficients will remain small and contribute minimally to the regression model. L2 regularization prevents overfitting of data while including all features within the model so that each feature can contribute to the prediction. Quantitative and qualitative data was included in the feature analysis, including the creature subtypes and ability text. To extract important words and phrases from ability text, TF-IDF and standard term frequency of the top 15 frequent words were included in the model. In addition, the most frequent subtypes, subtypes with prior knowledge of having higher prices through game knowledge, rarity, color type, and foil finishes were incorporated into the feature dimension using multiclass one-hot encoding. Due to the high dimensionality and incorporation of all card elements, the Ridge regression model was the best linear choice over Lasso Regression to analyze for this project.

To evaluate the coefficients and scores at each L2 lambda value, a for-loop of the ridge regression with a 5-fold cross validation split of training and test data is conducted for each lambda value. The output includes all scores and coefficients for each lambda value to feed into coefficient path plots and residual plots used analyzed to evaluate the model's performance (Plots can be found in Appendix B3-B6).

Supervised Learning - Overall Evaluation Metrics (Model Comparison)

We used mean absolute error, mean squared error, and r-squared scores to evaluate our models. R-squared is to measure variance, MAE the difference between min and max values, and MSE for the presence of heavy errors.

Table 1 - Supervised Evaluation Metrics

Model	Mean Absolute Error (MAE) <i>Mean (std dev)</i>	Mean Squared Error (MSE) <i>Mean (std dev)</i>	R-squared <i>Mean (std dev)</i>
K-Nearest Neighbors (Creatures)	4.15 (0.81)	215.76 (40.73)	0.05 (0.02)
K-Nearest Neighbors (Non-creatures)	9.00 (0.81)	2794.40 (339.09)	0.09 (0.06)
Random Forest (Creatures)	4.25 (0.18)	239.03 (32.52)	0.24 (0.03)
Random Forest (Non-creatures)	9.93 (0.49)	2425.00 (486.36)	0.19 (0.03)
Ridge Regression (Creatures)	13.55 (0.79)	6120.50 (2910.84)	0.03 (0.01)
Ridge Regression (Non-Creatures)	52.53 (6.58)	69132.69 (41602.69)	0.22 (0.11)

All values are rounded to the hundredth place.

As seen in Table 1, the highest R-squared value for the Creatures subsets was the Random Forest Regression at 0.24 ± 0.03 . However, the K-Nearest Neighbors has the lowest MAE and MSE scores for the creature dataset. Due to the MAE and MSE scores being similar between both the K-Nearest Neighbors and the Random Forest Models, the Random Forest Model was chosen as the best model due to having the larger R-squared score of the two models and having the smaller variance for the MAE and MSE scores. For the Non-creature subsets, the highest R-squared value is the Ridge regression model at 0.22 ± 0.11 . However, both the MAE and MSE are amongst the highest of all the models. The next best model is the Random Forest model at 0.19 ± 0.03 . Between K-Nearest Neighbors and the Random Forest

Understanding Magic: the Gathering Card Value through Machine Learning

Sophia Chang, Janel Jackson, Jessica Lan

models, the Random Forest Model provides the lowest MAE and MSE scores and the highest R-square score, making it the best model out of the three analyzed.

Given the Ridge regression had the worst performance amongst most scores, it is extremely likely that the dataset does not correlate linearly with market price. Given there are a significant number of features derived from one-hot encoding of card sub types, rarity, and foil finishes in addition to tf-idf values from the ability text included in both the Random Forest and KNN models, the Random Forest outperformed the KNN model. Random Forest Model also prevents overfitting due to the randomness nature of the tree formation and thresholds formed at each iteration. Even though the Random Forest Model has been chosen as the best model out of the 3 within the analysis, the highest R-squared scores out of all 3 models are still significantly low for both creature and non-creature sub types.

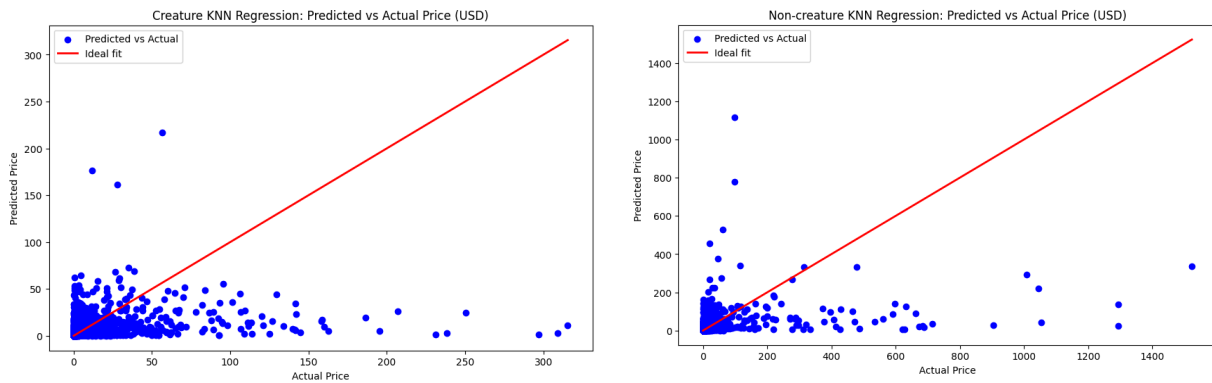


Figure 1 - K-Nearest Neighbors regression on creature vs. non-creature cards.

The output for the KNN model suggests all cards are predicted to be below 50 dollars in the creature model while the majority of cards are below 100 dollars and spread up to 350 dollars. For the non-creature model, the vast majority of cards are predicted to be below 800 dollars while they are actually below 200 dollars and span up to 1200 dollars. The model tends to estimate too high for the low cost cards and vice versa, estimating prices at around 0-50 dollars for creature cards and 0-400 dollars for non-creature cards. This may be because of how high some of the actual card prices are even after processing for price outliers using z-score and grouping together cards with the same name but different price listings due to different, limited card art (price is averaged), and cards with the same name but different traits like card finishes are kept in separate rows). There is an undue right skew where the number of cheap cards outnumber the expensive ones, but the expensive cards make the model not fit well with the rest of the cheap cards.

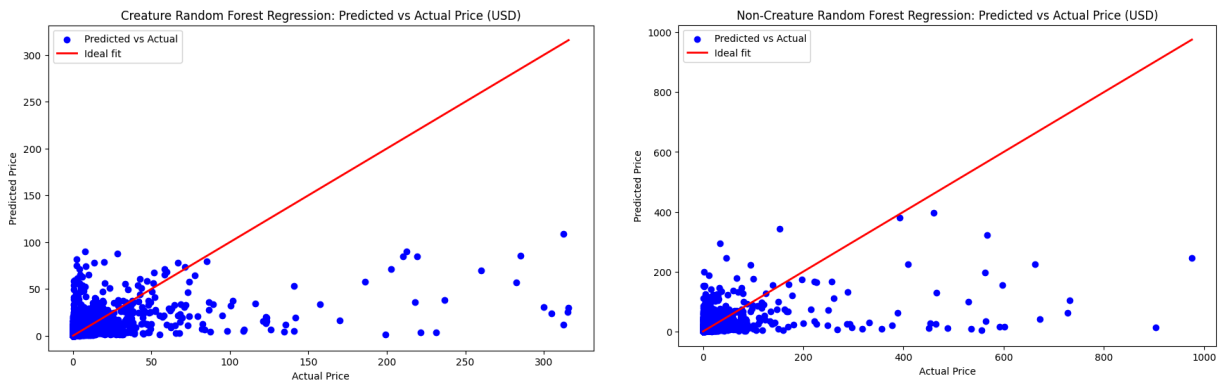


Figure 2 - Random forest regression on creature vs. non-creature cards

Understanding Magic: the Gathering Card Value through Machine Learning

Sophia Chang, Janel Jackson, Jessica Lan

The random forest regression results show a wider spread of predicted values compared to KNN. For creature cards, most predictions fall between 0 to 100 dollars while for non-creature cards, predictions extend up to 400 dollars. There is significantly more variance in the non-creature model than the creature model, likely due to the broader price range predictions and the actual wider distribution of non-creature card prices. Some points lie close to the trendline, indicating an accurate price estimate for certain cards. However, for both creatures and non-creature, the model performs poorly on higher priced cards. As with the KNN model, this may be due to the right skewed price distribution in our dataset, where lower priced cards dominate the dataset and expensive cards are relatively rare, limiting the model's ability to learn patterns from the high-value cards.

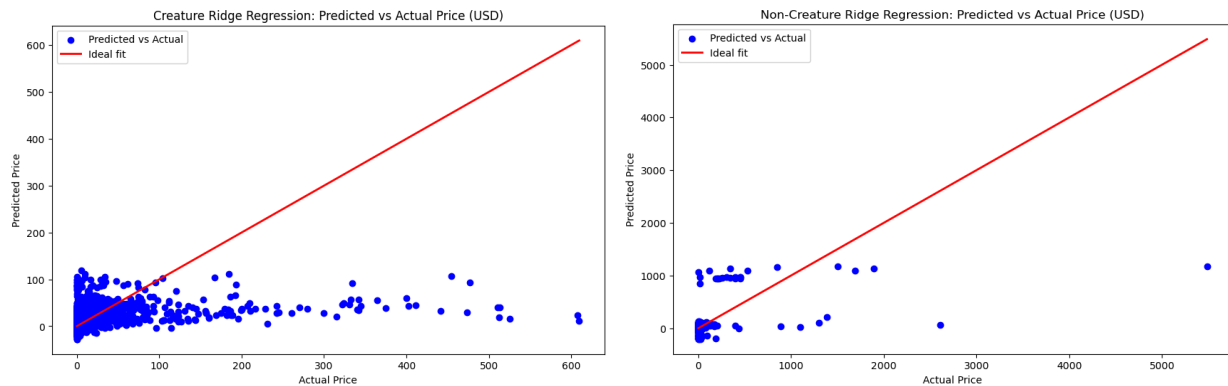


Figure 3 - Ridge Regression on creature vs. non-creature cards

For the ridge regression model, the creature model output is also restricted to estimating prices at maximum of 100 dollars. However, as the actual price of the card increases, the minimum estimated price also increases which can be seen by the number of data points drifting away from the predicted cost of 0 dollars as the actual price increases. The non-creature model shows estimated prices at either 1000 dollars or at 0 dollars. This could be due to the largest coefficient ('isReserved') being equal to \$839.55 USD and this field being binary. It is safe to assume that all of the cards with an estimated price of approximately 1000 dollars are Reserved cards. Reserved cards are cards that have been confirmed will never be reprinted again due to retaining value in rare, old and iconic cards. This field will be analyzed in the failure analysis section of the report to further investigate potentially why the gap exists.

Supervised Learning - Best Model Evaluation Metrics

Feature Importance

For the creature and non-creature card types, the random forest regression model achieved the best performance, with an R-squared score of 0.24 and 0.19 respectively. The random forest regression model provides a measure of feature importance as part of its output. Based on the feature importance plots below (Figure 4 & 5), the top important features for the creature model are 'edhrecRank', 'edhrecSaltiness', 'isPromo', 'isAlternative', and 'isMTGO' while the top features for the non-creature model are 'edhrecRank', 'edhrecSaltiness', 'isReserved', 'manaValue', and 'isGameChanger'.

Understanding Magic: the Gathering Card Value through Machine Learning

Sophia Chang, Janel Jackson, Jessica Lan

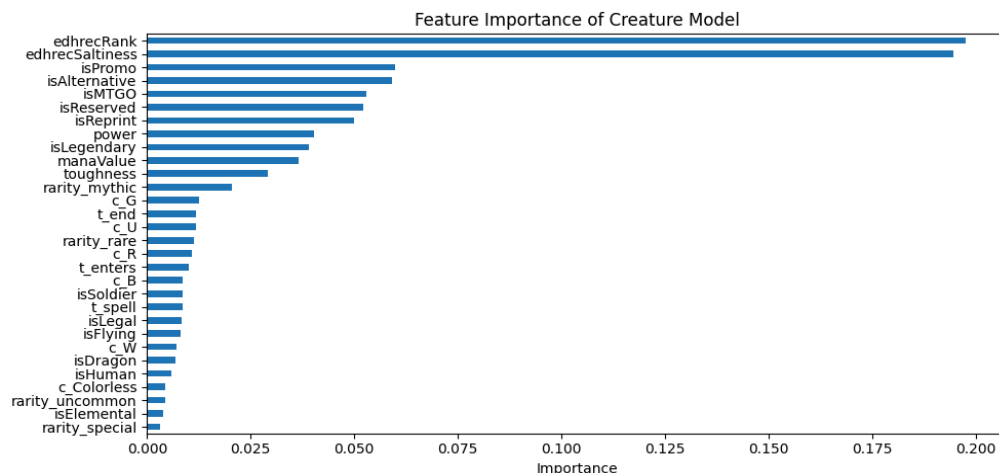


Figure 4 - Feature importance for random forest regression on creature cards, top 30 of 52 features

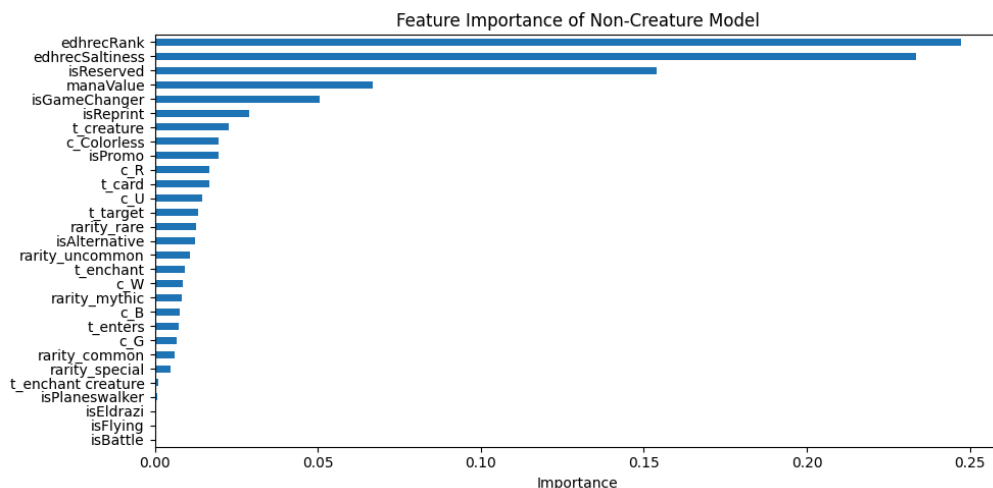


Figure 5 - Feature importance for random forest regression on non-creature cards

The feature importance results largely align with expectations for card market prices. For both creature and non-creature cards, 'edhrecRank' and 'edhrecSaltiness' being among the most important features makes sense since customer-driven features like popularity and saltiness can drive up the demand and price of a card. Highly "salty" cards tend to be powerful or frustrating to play against, which corresponds to strong gameplay impact and thus higher demand. Notably, these customer-driven factors rank higher than card-driven attributes such as power, toughness, and mana value. For creatures, the importance of 'isPromo' and 'isAlternative' suggests that special edition cards and print variations can influence price. Promotional cards are not found in regular sets and are obtained through prerelease events, giveaways, or other special events. Since these cards are limited in quantity, they are typically more expensive due to scarcity. The 'isAlternative' feature refers to alternative print variations such as unique frames or artwork variations. These versions are typically more rare than standard printings, which may appeal to collectors and increase their demand.

A similar explanation applies to the 'isReserved' feature for non-creature cards. Cards on the reserved list are guaranteed to never be reprinted, and as such, these cards are more expensive due to their limited supply. Additionally, 'isGameChanger' refers to a curated list of over 40 cards that Wizards of the Coast identifies as high-power for Commander deck construction. The importance of this feature suggests that cards with overwhelming advantage in gameplay are priced at a premium. The inclusion of cards on the list can also increase demand. While these feature importance results are intuitive, the low R-squared

Understanding Magic: the Gathering Card Value through Machine Learning
Sophia Chang, Janel Jackson, Jessica Lan

score indicates that a large portion of the price variance remains unexplained, likely due to missing variables in our dataset.

When performing an ablation analysis on the random forest regression creature and non-creature model by removing top features and retraining the model, we observed the following impact on model performance. The drop in R-squared score for both the creature and non-creature model reiterates the importance of the customer-driven features, popularity and saltiness, in predicting card market price.

Table 2 - Ablation Analysis Metrics

Model	Mean Absolute Error (MAE)	Mean Squared Error (MSE)	R-squared
Random Forest (Creatures) - drop 'edhrecRank'	4.50 (0.16)	242.79 (31.59)	0.22 (0.03)
Random Forest (Creatures) - drop 'edhrecRank' & 'edhrecSaltiness'	4.80 (0.14)	256.39 (26.35)	0.18 (0.03)
Random Forest (Non-creatures) - drop 'edhrecRank'	10.37 (0.47)	2504.96 (523.51)	0.17 (0.03)
Random Forest (Non-creatures) - drop 'edhrecRank' & 'edhrecSaltiness'	11.13 (0.47)	2600.66 (523.17)	0.14 (0.03)

All values are rounded to the hundredth place.

Sensitivity Analysis

We ran a sensitivity analysis on the random forest regression model to evaluate how model performance is affected by the choice of hyperparameters. The best performing creature model used the hyperparameters {'max_depth': 15, 'max_features': None, 'min_samples_leaf': 5, 'n_estimators': 100}, and achieved an R-squared score 0.24. In comparison, a model configured with {'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 3, 'n_estimators': 50} achieved an R-squared score of 0.15. This 0.09 decrease in R-squared suggests that model performance is moderately sensitive to hyperparameter selection. In particular, increasing the number of estimators and the maximum tree depth improves model performance, due to reduced variance and improved ensemble stability.

In contrast, performing the same sensitivity analysis on the non-creature model did not impact model performance. Using the alternative hyperparameter configuration resulted in the same R-squared score of 0.19. This may be because non-creature pricing is driven by a smaller set of key predictors, making the model simpler and less dependent on tree depth or ensemble size to capture the underlying pricing relationship. In comparison, the creature model likely involves more complex feature interactions, requiring greater model capacity to achieve optimal performance.

Tradeoffs

The main tradeoff for the random forest regression model is speed versus accuracy. By increasing the number of decision trees and maximum tree depth, accuracy improves because the model is able to capture more complex patterns in the data, however this comes with the cost of longer training time and increased memory usage. Another tradeoff is model complexity versus interpretability. As the number of decision trees and tree depth increases, the model also becomes more difficult to interpret. The best performing model uses 100 decision trees each with a maximum depth of 15, and the final price

prediction is computed as the average across all trees. This makes it more difficult to explain why a certain price was predicted, since it would require looking at multiple deep trees.

Supervised Learning - Failure Analysis

For K-nearest neighbors regression, there was one value in the creature model that had a predicted price of 116 USD whereas the actual price was 4 USD. It was a rare card with a foil finish and white mana, was a human with the flying keyword, contained “end” in the text, had middling power, toughness, and mana value, and wasn’t particularly popular or infamous, with a #5635 rank and 0.53 saltiness on EDHREC). This card is [Spectacular Spider-Man](#). Possible reasons why this card got a disproportionately high score are the keywords and possibly the foil, but they weren’t particularly important features. This card only had 256 foil printings (the rest are nonfoil), and the character the card depicts is one of the most popular superheroes in the Marvel franchise, but our model doesn’t account for crossover appeal (and if it did, that new model would probably predict it to be more expensive). It may be that there is a systematic error in the model itself as KNN on creatures did particularly poorly and this one card is a prominent example, or an operational issue with regards to data quality after preprocessing.

For random forest regression, the non-creature model predicted a card worth 388.38 USD as only 62.93 USD. This card is [Deathlace](#), an Instant card type with ranking of 28376 and saltiness of 0.31 on EDHREC, rarity type of rare, mana value of 1, and containing the word ‘target’ in its ability text. In comparison, the model was able to predict another card in a similar price range more accurately. The [Lion’s Eye Diamond](#) card was predicted as 380.72 USD, fairly close to its actual price of 391.94 USD. This card is an Artifact with ranking of 744, saltiness of 1.54, rarity type of rare, and mana value of 0. This card is also on both the game changers list and reserved list. Given that ‘ehrecRank’, ‘edhrecSaltiness’, ‘isReserved’, ‘manaValue’, and ‘isGameChanger’ are the five most important features, it is reasonable that the model was able to more accurately predict the Lion’s Eye Diamond card. The DeathLace card has a much lower popularity ranking (high number means lower rank), lower saltiness (on scale of 0-4), and higher mana value, meaning the card requires more cost to play. An attribute not captured in our existing features that could explain the prediction error is that this card is in the ‘Limited Edition Alpha’ set. This set was released in 1993 as the very first printing of MTG with a limited print run (roughly 1,100 of each rare card). As the first edition, these cards are very sought after by collectors, making them extremely expensive. In comparison, the Deathlace ‘Fourth Edition’ card is only worth 35 cents.

This discrepancy indicates a systematic error in our models which does not capture set level value. As a result, the model performs well when prices are driven by gameplay signals, but underperforms when prices are driven by edition rarity. To improve our model, we should incorporate set-level features, either by one-hot encoding the set name (due to vast number of sets, this may not be the best approach), or manually creating boolean features for sets that may be associated with higher prices (e.g. isLimitedEditionAlpha, isLimitedEditionBeta). We should also explore if there’s other factors not in our existing datasets that can explain the set prices. Finally, since these cards differ in card type (Instant vs. Artifact), we should evaluate whether certain non-creature types should be modeled separately.

As a result of the non-creature ridge regression model (the right graph in Figure 3), there is a gap in the scatter plot where cards fall either around \$1000 or \$0 predicted market value. When reviewing the coefficients within the non-creature model, the “isReserved” feature has a coefficient of \$839.55 while the next largest feature (“isLegal”) has a coefficient of only \$197.95. Given there is such a large gap between “isReserved” and “isLegal”, a SHapley Additive exPlanations (SHAP) analysis was conducted to understand how much of an impact cards on the Reserved list impacted the test results. When looking at the card “Tropical Island” at the time of running the analysis, the actual market value price was \$5,485.35. “Tropical Island” is one of the few cards in the dataset on the Reserved List, which contributed to the output of \$1,193.97 (Seen in Appendix B7). The results reflected that the feature with “isReserved” marked True added an additional \$892.69 to the prediction. It may be worth removing these Reserved list cards or the “isReserved” feature from the model given most of these cards have a special condition compared to all of the other cards given that they will never be reprinted again. It seems to be introducing bias into the model, especially given that future cards that would be added to this dataset would not be considered a Reserved list card.

Unsupervised Learning - Models Analyzed

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Uniform Manifold Approximation and Projection (UMAP) clustering models were the unsupervised learning methods used to evaluate patterns in price, subtype, keywords, color identity, and artist/art style via dimensionality reduction and clustering. Only creatures were included in the dataset, given the most amount of variety in the artwork is within the creature dataset.

DBSCAN is a non-parametric clustering algorithm. It determines clusters by grouping points that are very close together and specifically marks outliers as their own label. It is this latter property that made us choose DBSCAN as one of our models. For DBSCAN, hyperparameter tuning was done through making a graph of the ideal epsilon score from nearest neighbors, then using a subset of viable epsilon values in a DBSCAN model and evaluating the models' silhouette scores to find the best performer. The number of minimum samples was acquired via the same process, except instead of using silhouette score, the minimum sample size was determined through how many clusters were there - ideally below nine but above three clusters.

UMAP is a non-linear, non-deterministic cluster model that takes in distances of features, identifies the best dimension to display a reduced number of features, and projects the new features onto a graph. Given the development of clusters is essential to compare similar cards with different market price values, UMAP considers both local and global cluster structures making it great for our analysis. To identify how well the clusters are formed, the x and y positions acquired from the UMAP classifier are incorporated into a K-Means clustering algorithm. A GridSearch cross validation was integrated into a pipeline to hypertune parameters for both UMAP and K-means algorithms and identify the best clusters based on clustering evaluation metrics.

Unsupervised Learning - Overall Model Evaluation and Comparison

The evaluation metrics used to identify the best model were silhouette score to measure similarity between a point and its cluster, Davies-Bouldin index for cluster-cluster similarity, and Calinski-Harabasz index for cluster dispersion. In addition, information about the artwork and market value price was further evaluated with the best model.

DBSCAN

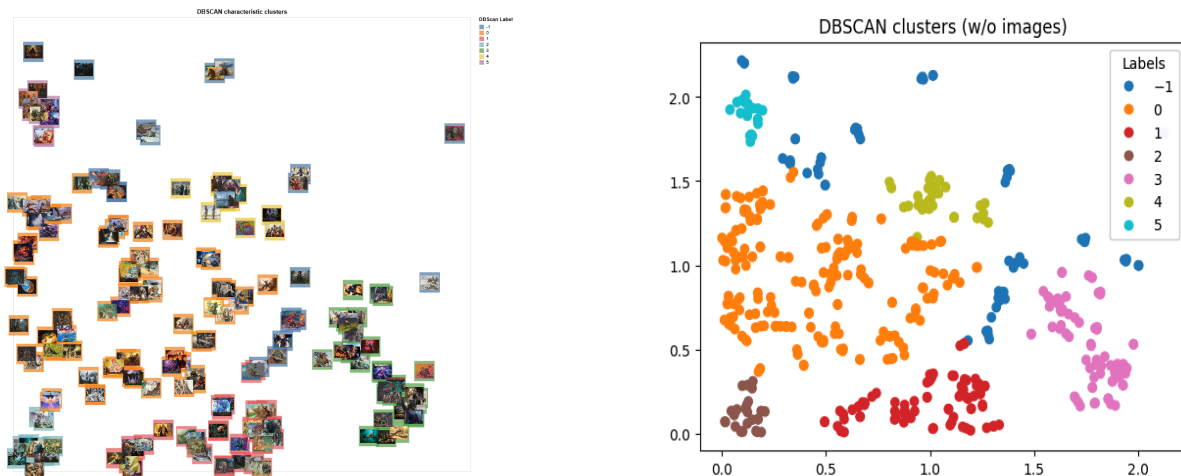


Figure 6 - DBSCAN clusters by card features - one with card art (left) and one without (right)

For the DBSCAN model, the silhouette score was 0.66 ± 0.11 , the Davies-Bouldin was 0.46 ± 0.22 , and Calinski-Harabasz was 1606.89 ± 2224.08 . All values were rounded to the hundredth place. These scores

indicate reasonably confident cluster cohesion, low cluster similarity, and disproportionately high cluster dispersion.

Most of the nonhuman cards were in cluster label 3 (bottom right), though with varying price numbers, none with a consistent trend. Many of the cards with the flying keyword were in cluster label 2 (bottom left), and most of them had a price of over 10 USD. Cluster label 4 (the small clump, top-center) contains majority-expensive cards of mostly blue color identity. Cluster label 5 (top left clump) is nearly-all human, and most of them had a price of at least 50 USD. Cluster label 1 (bottom-middle) had no discernable price or keyword trends, other than being made up of humans or animals with the odd decidedly-fantasy race here and there. The most numerous and wide cluster, label 0 (large collection spread out from middle-left to middle) contains a variety of low and high-cost cards and traits that mostly conform to what is expected - humans are common, and by themselves don't add value, especially if they're 'merely' a lowly commoner or soldier, while more exotic subtypes like goblins are rarer and may be more valuable. Common mismatches from the cards classified as noisy (cluster label -1, scattered points on the upper side and between clusters 0 and 3 of the graph making a sort of 85-90 degree angle) are from the model predicting high prices for low-priced cards that had desirable traits like flying or multiple color identities, or from a limited edition (compare this [60 USD printing](#) vs. the normal [1.23 USD printing](#) of Cloud, Ex-SOLDIER) of a specific card. There were no discernable insights when trying to look at artists (while there were instances of cluster-artist overlap, it is a minor occurrence - only 2 cards from the same artist in a cluster).

UMAP with K-Means Clustering

The parameters adjusted specifically for the UMAP classifier included the quantity of neighbors within a cluster, the minimum distance between points within a cluster, and the metric used to calculate distance. The only parameter that was tuned was the number of clusters for the K-mean model. As a result, the UMAP visualization using K-means clustering verified that a total number of 6 clusters, with a distance of 0.1 and 20 neighbors per cluster produced the best silhouette score, Davies-Bouldin score, and Calinski-Harabasz score, all calculated to be 0.66 ± 0.02 , 0.49 ± 0.02 , and 1229.92 ± 215.47 respectively.

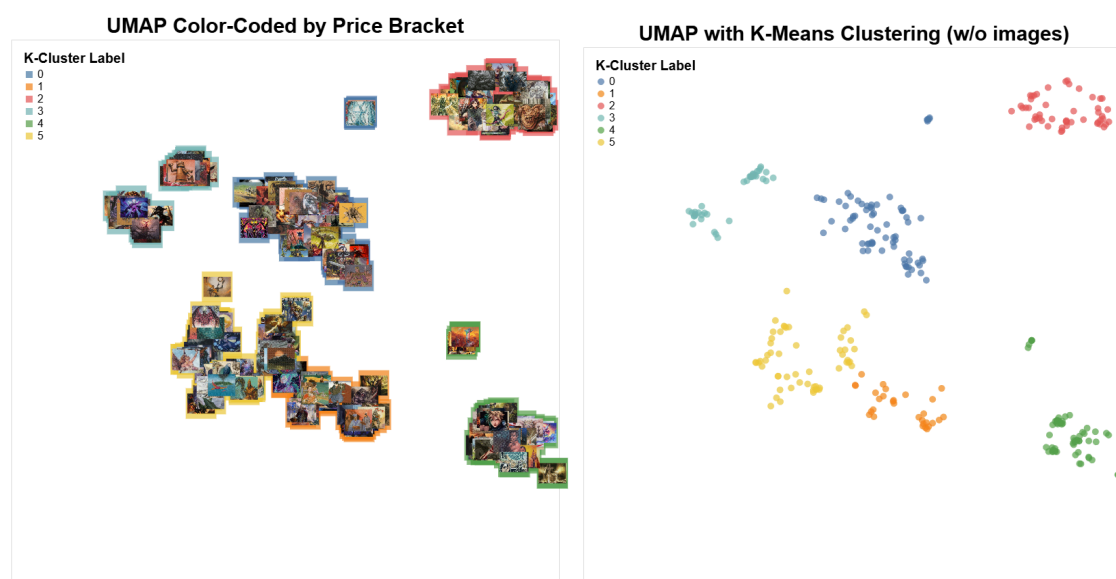


Figure 7 - Visualizations of UMAP using K-Means Clustering Algorithm color-coded with (L) and without (R) images

In Figure 7, each cluster mostly has been separated by one to two color types except for the last grouping that mostly have flying creatures. And within each cluster, the groups are then subdivided into prominent creature subtypes. Regarding artwork, the clusters are separated by color type (Clusters 0-4) reflect the color identity within the art of the card. For example, black cards tend to be really dark and there doesn't

Understanding Magic: the Gathering Card Value through Machine Learning
Sophia Chang, Janel Jackson, Jessica Lan

appear to be much white to them unless it is to highlight a very dark object or creature within the art. However, red creatures have a lot of red elements to them, the scenery includes red, the creatures are red, or there are fire elements within the art pieces. For Cluster 5, the artwork incorporates a creature with wings, flying through specific terrain and usually incorporates a forest or sky in the setting. Below is Table 3 describing prominent and consistent card details identified within each cluster:

Table 3 - K-Means Cluster Descriptions for UMAP Visualizations

Cluster ID	Primary Color Identity of Creature Cards	Main Creatures Types within Cluster	Subtypes of Groupings within Clusters
0	Red	Goblin, Human, Lizard, Beasts	Green, Red, Blue, and White creatures
1	Blue only and Black only	Human	Wizards, Rogues, Warrior
2	Green	Elves, Humans, Plants, Dinosaurs	Druid, Warrior, Archer
3	Colorless only and Black Creatures	Golem, Construct	Phyrexian
4	White only and Multicolor containing White	Human, Soldier	Cleric, All Color (White, Black, Red, Green, and Blue)
5	All Colors	Flying, Elemental	Bird, Dragon, Spirit, Angel

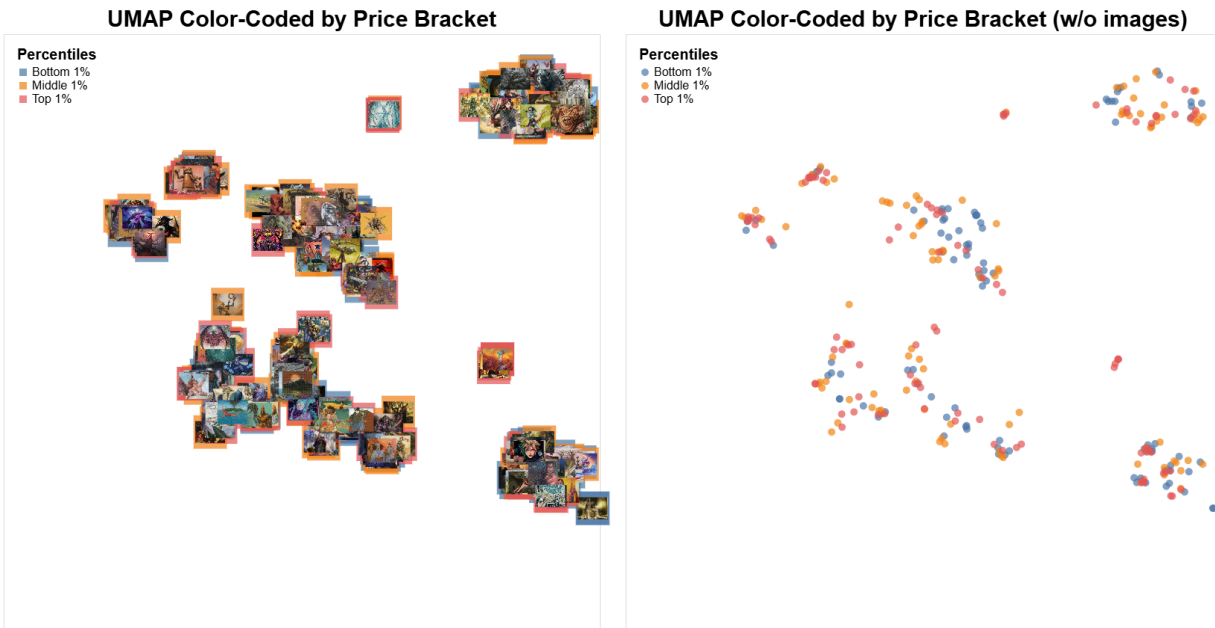


Figure 8 - Visualization of UMAP clusters with (L) and without (R) images (clusters are top, bottom, median 1%)

Understanding Magic: the Gathering Card Value through Machine Learning

Sophia Chang, Janel Jackson, Jessica Lan

In Figure 8, there are 100 randomly selected images within each percentile bracket to make it easier to distinguish artwork within each cluster. The top 1% bracket seems to contain painted artwork, artwork with an anime style, or artwork that contains figures from pop culture references. For example, Cloud from Final Fantasy appears in the top left in Figure 10 twice and are both incorporated in the Top 1% bracket. Gandolf from Lord of the Rings can also be seen waving his staff at the bottom center of Figure 9 as well. Some of the Top 1% cards have a traditional fantasy style painting aesthetic to them reminiscent of mainstream fantasy games that were popular in the 1990s, like Dungeons and Dragons. This can be seen with the bottom right cluster in Figure 10 with the image of a bear and the man in green. This suggests that vintage cards and cards that cross over different media types may increase the market value of a card. However, there doesn't seem to be a difference between the Middle 1% bracket and the Bottom 1% bracket. To confirm if vintage cards and pop culture reference cards have an influence on market value, the set and release date could be integrated into future models, which may alter the clusters more towards the expected price brackets.



Figure 9 - Images of K-Means Cluster 4 with price color-coding



Figure 10 - Images of K-Means Cluster 2 with price color-coding

Unsupervised Learning - Best Model Evaluation Metrics

Table 4 - Unsupervised Evaluation Metrics Comparison

Model	Silhouette Score (std dev)	Davies-Bouldin Score (std dev)	Calinski-Harabasz Score (std dev)
DBSCAN	0.66 (0.11)	0.46 (0.22)	1606.89 (2224.08)
UMAP with K-means Clustering	0.67 (0.02)	0.49 (0.02)	1229.92 (215.47)

All values are rounded to the hundredth place.

According to Table 4, both DBSCAN and UMAP with K-Means clustering both have similar silhouette and Davies-Bouldin scores. However, the UMAP model Calinski-Harabasz Score is slightly lower than the DBSCAN Score indicating that the intercluster difference is high with clusters being spaced out well while the intracluster difference is low indicating clusters are tightly formed together for both models. However with such the large variances of the DBSCAN, the UMAP with K-means clustering provides a more consistent performance and has been selected as the best model to evaluate hypertuning.

There are several hyperparameters with the UMAP manifold learning algorithm and one main hyperparameter for the K-means clustering. Given the labels of the clusters drives the evaluation metric scores, the number of clusters was adjusted to identify how impactful changing the quantity of clusters is to the performance of the model.

Table 5 - K-means Clustering Cross Validation Scores (3 Train/Test Split)

Number of Clusters	Silhouette Score <i>Mean (Std Dev)</i>	Davies-Bouldin Score <i>Mean (Std Dev)</i>	Calinski-Harabasz Score <i>Mean (Std Dev)</i>
4	0.64 (0.03)	0.56 (0.05)	720.60 (85.46)
5	0.68 (0.03)	0.45 (0.04)	994.10 (182.50)
6 (Best Performing Model)	0.67 (0.02)	0.49 (0.02)	1227.69 (213.89)
7	0.63 (0.01)	0.51 (0.08)	1115.82 (146.02)
8	0.58 (0.02)	0.63 (0.06)	1141.59 (206.10)

All values are rounded to the hundredth place.

Based on the scores in Table 5, the lowest silhouette score peaks at 5 and 6 clusters, with the 6 clusters having a smaller mean silhouette score, but a smaller variance making it a bit more concise than the 5 cluster model. This can mean the clusters are slightly more similar to each other in the 5 cluster model than the 6 cluster model. However, the lowest value is at 0.58 with 8 clusters, but given the high number of clusters, the labels may be overfitting the dataset. When reviewing the Davies-Bouldin Score, the same trend occurs but with the 5 cluster model having a slightly smaller mean Davies-Bouldin Score but a higher variance than the 6 cluster model. In this case, the 8 cluster model has the highest Davies-Bouldin score, which can confirm that 8 clusters may not be the optimum quantity of clusters for the dataset. Finally, when looking at the Calinski-Harabasz Score, the average score for the 6 cluster model outperforms all other models even though the variance is the highest. To really understand the impact of the 5 cluster model versus the 6 cluster model, the clusters within each visualization are compared to confirm 5 or 6 clusters make more sense in Figures 11 and 12 below.



Figure 11 - Clusters formed with 6 clusters

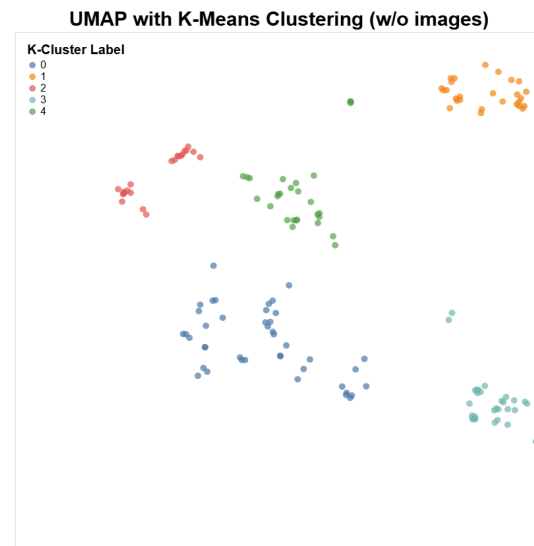


Figure 12 - Cluster formed with 5 clusters

The yellow and orange cluster in Figure 11 combined to create the large blue cluster in Figure 12 when reducing from 6 clusters to 5 clusters. The combined dark blue cluster has a more oblong shape compared to the two separated clusters that are almost split in half. Based on this, the cluster that was newly formed separated the flying creatures in yellow with the non-flying creatures in orange. Due to the improved spherical shape of the clusters and the clear labels, the separation of the large dark blue cluster performed significantly better than them combined making the 6 cluster model the preference over the 5 cluster model.

Conclusion

Supervised Learning

What surprised us was that there weren't any significant trends we could identify in our models. Much of our work was stymied by our metric scores indicating that many of our models were poor fits for the data. Additionally, we realized that there were many listings for slightly different versions of the same card, with the same power/toughness/rank/saltiness but different finishes and, most importantly, price. We changed our pre-processing methods, hyperparameter tuning, added different (and removed unimportant) features, and consolidated card listings and averaging price, though there was only minor improvement. However, according to our failure analysis, set information may be an important feature to include in future analyses, and Reserve cards, cards that will never be reprinted, may have introduced bias into our models. With more time, we would recommend excluding Reserved Cards from our dataset, searching for additional datasets with extra set information, and incorporating all/most keywords in the card listings.

Unsupervised Learning

Based on the UMAP model, color identity and creature subtype were the primary focus of the clusters for the creature dataset. When adding artwork in for points on the graph, the pricier cards mainly contained antique cards with hand paintings of fantasy from the 1990's, cards with a distinct Japanese anime art style, or incorporated art from pop culture tv shows, video games, and movies. It is very likely that with the inclusion of release date of the card and the set the cards were released in should be included as features within the algorithm. This ties back to supervised learning findings where set information should definitely be analyzed with the rest of the dataset to provide more clarity about the market value of the card. With this information added to the visualizations above, the clustering may provide even more insight as to what additional features can be extracted through observation.

Ethical Considerations

Some people may use these findings to mass buy cards and resell them at artificially high prices for profit, but it is likely they were going to do that anyway with or without the presence of our report. Speculative markets are extremely volatile in general, but that is an existing issue. In addition, the model must be trained on price data post release of new cards to ensure market stability. If the model incorporates prices of cards immediately after cards are revealed through official social media sites or through websites but before they are released for sale, the prices are extremely inflated due to speculation, pre-sale bidding of popular cards, and other market manipulations. Given the prediction model is already underperforming, trying to predict pricing right after the announcement of a new release of cards will make the predictions of market value price even more inaccurate.

Statement of Work

Janel Jackson worked on ridge regression and t-SNE/UMAP K-means clustering. Jessica Lan worked on random forest regression and MDS artwork clustering. Sophia Chang worked on the k-nearest neighbors and DBSCAN clustering.

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Understanding Magic: the Gathering Card Value through Machine Learning
Sophia Chang, Janel Jackson, Jessica Lan

Appendix A: Final Supervised Learning Features

- power (creature-only, attack power)
- toughness (creature-only, defense)
- edhrecRank (popularity in EDHREC)
- edhrecSaltiness (a measure of how annoying this card is to play against from EDHREC users)
- manaValue (cost to play a card)
- one-hot-encoded color identities (colorless, blue, etc.)
- isAlternative
- isReprint (whether this card is a newer reprint of an older one)
- isReserved (cards that Wizards of the Coast say will never be reprinted)
- isPromo
- a variety of subtype features for creatures (isHuman, isEldrazi, isCleric, etc.)
- card availability (only in MTG Online, in physical, in MTG Arena)
- card finish (etched, foil, etc.)
- card rarity (common, uncommon, rare, mythic)
- frequently appearing words from the card's ability text ("enters", "end", "flying", etc.)
- card legality in commander mode
- loyalty (planeswalker-only)

Appendix B: Additional Visualizations

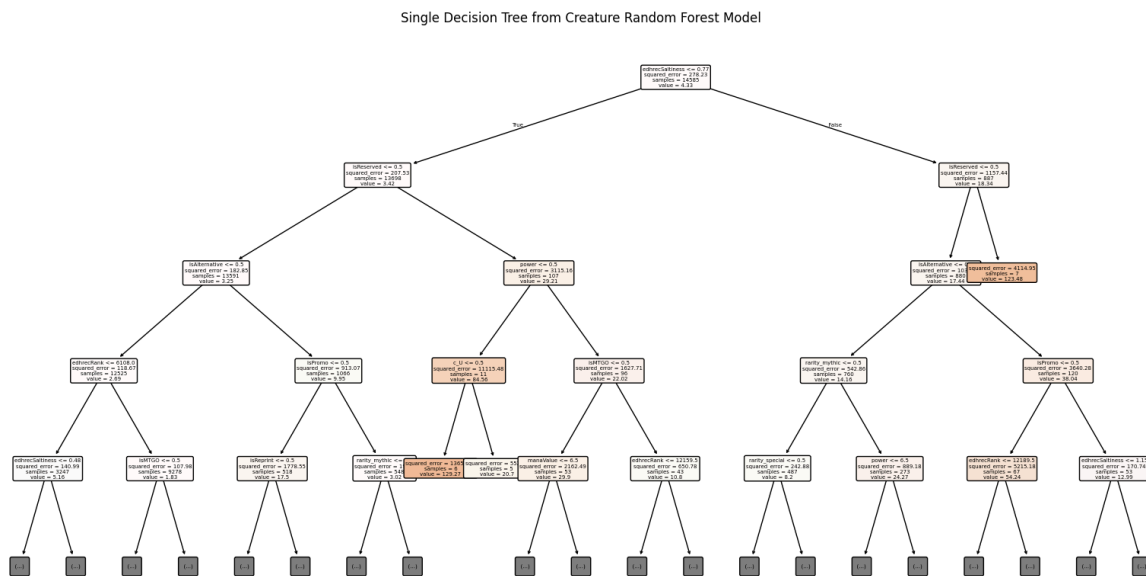


Figure B1 - Top 5 layers of 15-layer decision tree for creature random forest regression

Understanding Magic: the Gathering Card Value through Machine Learning
Sophia Chang, Janel Jackson, Jessica Lan

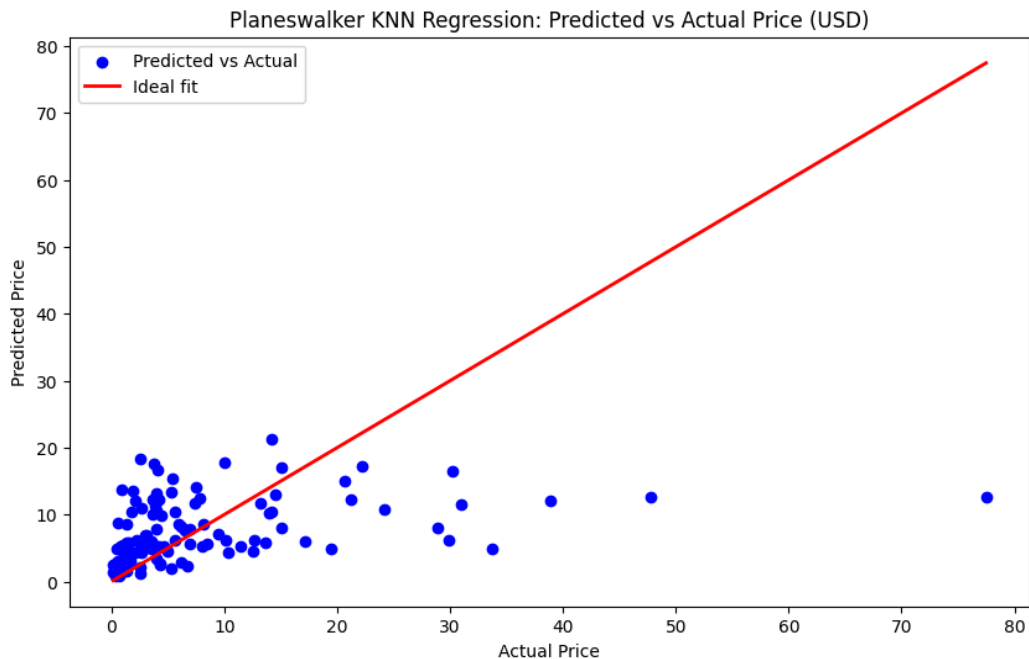


Figure B2 - K-Nearest Neighbors regression on *Planeswalker* cards. Omitted due to comparative lack of size.

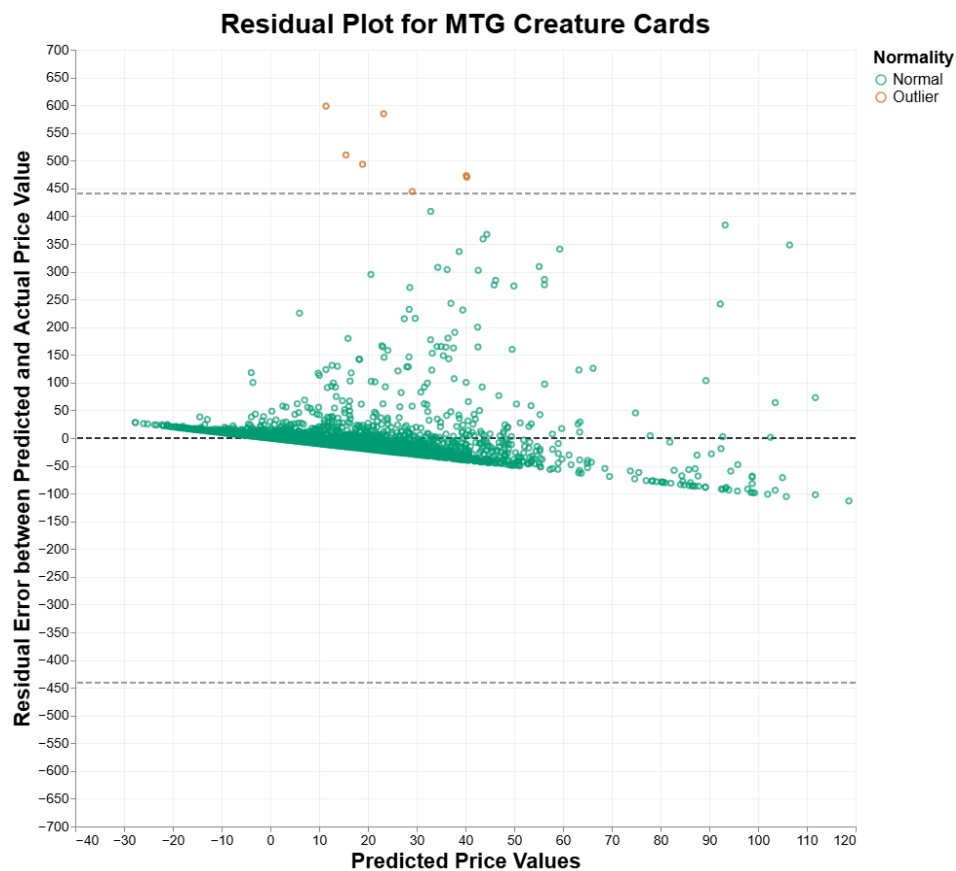
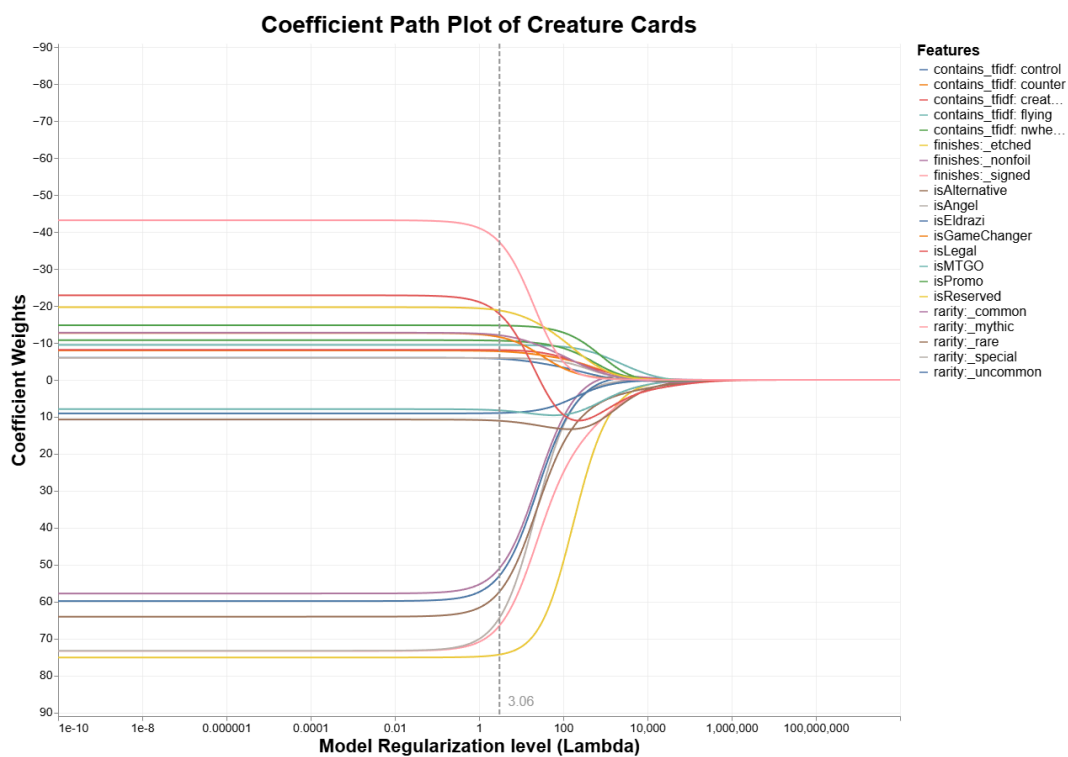
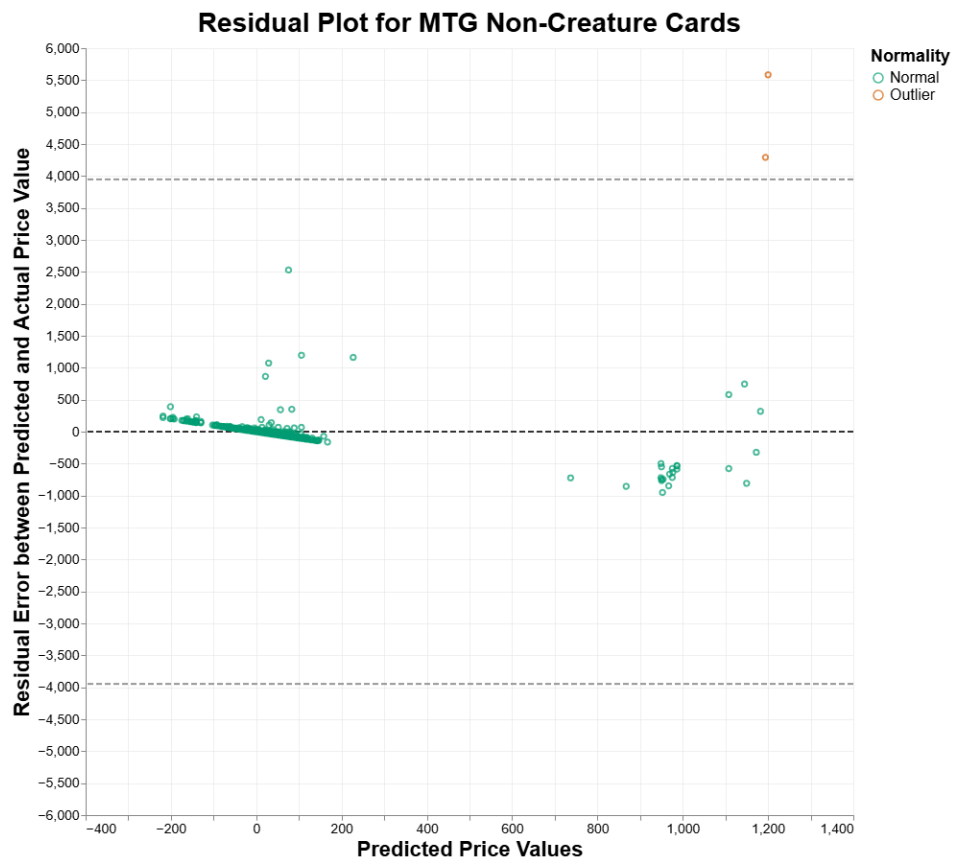


Figure B3 - Ridge Regression Residual Plots for Creature Dataset



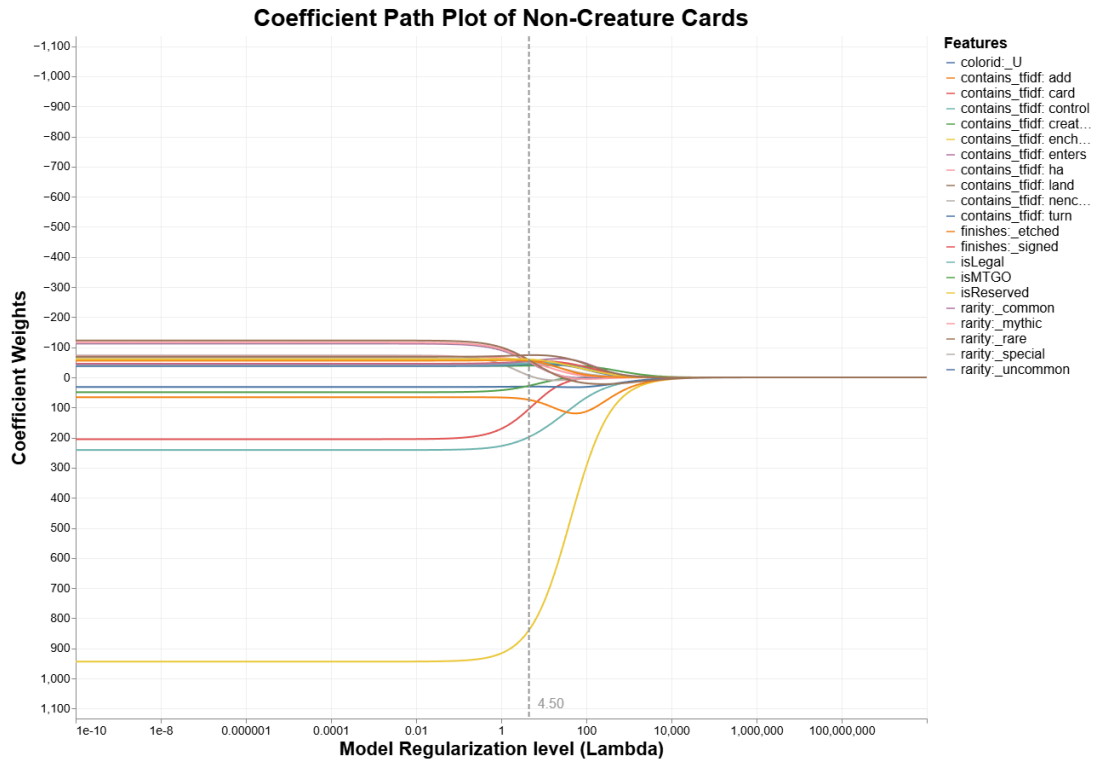


Figure B6 - Coefficient Path Plot for Non-Creature Dataset

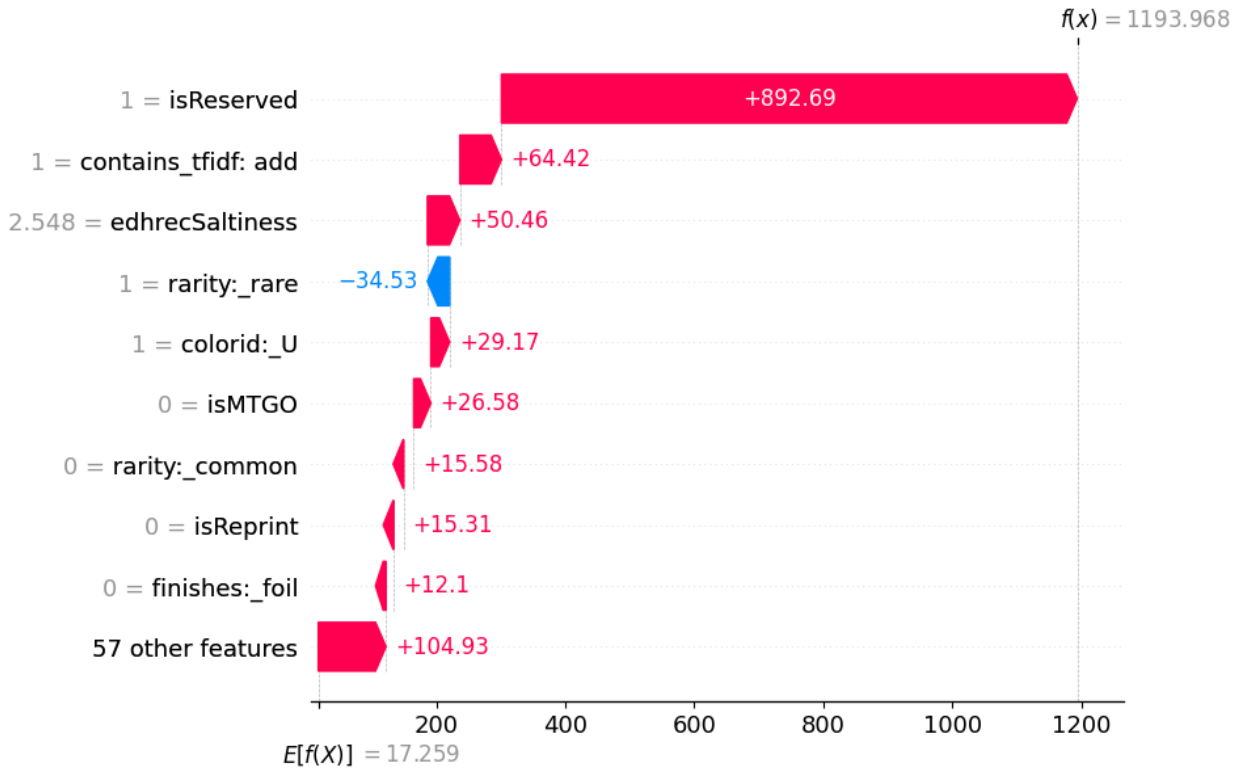


Figure B7 - SHAP Feature Importance Analysis of Tropical Island Land Card for Failure Analysis

Understanding Magic: the Gathering Card Value through Machine Learning

Sophia Chang, Janel Jackson, Jessica Lan

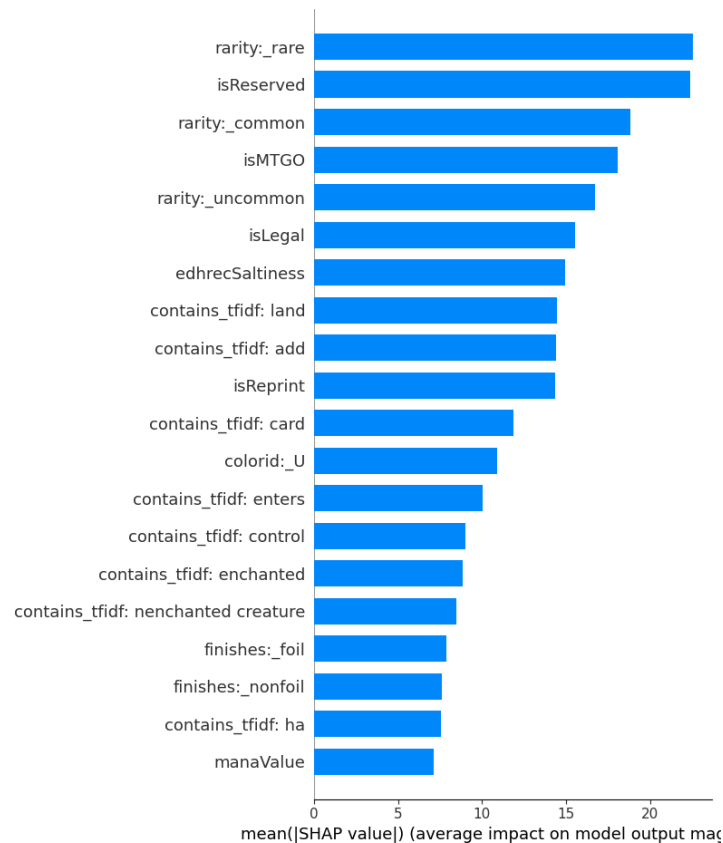


Figure B8 - SHAP Feature Importance Analysis of Test Data Set for Failure Analysis

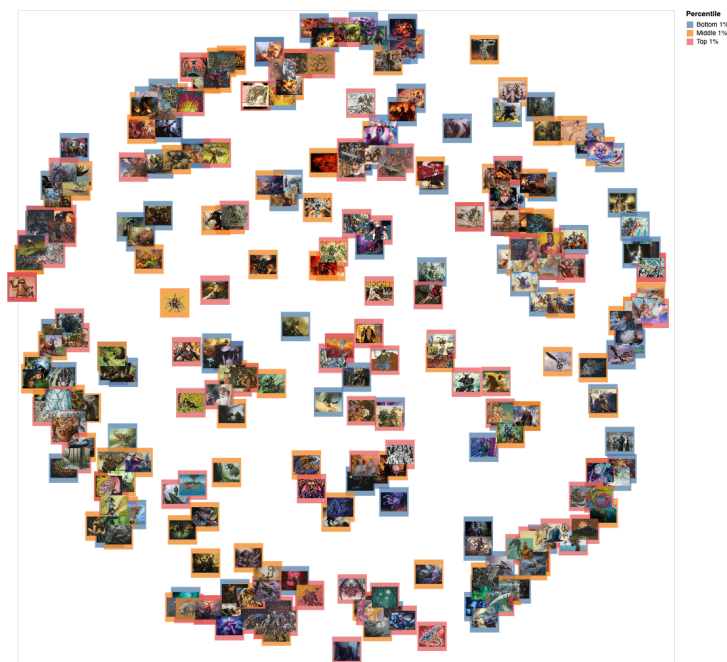


Figure B9 - MDS clusters, color-coded by price bracket (top, bottom, median 1%). Omitted due to poor clustering results compared to DBSCAN & UMAP